# NAUFAL BAGAS GANEFRIADY

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## SUMMARY

Final-year Electronics & Instrumentation student (GPA 3.88/4.00, Universitas Gadjah Mada) recently completing a Digital Transformation internship at PT Pertamina Hulu Rokan — Indonesia's largest upstream oil & gas producing block. Holds IWCF Level 1 Well Control certification. Research background includes fault-tolerant industrial IoT communication for mining safety-critical environments and Vision Transformer benchmarking on Earth observation data at Waseda University, Tokyo. Demonstrated ability to lead complex, multi-disciplinary engineering programs: directed an 8+ project, 50-person engineering organisation. Targeting early-career roles in digital technology, electronics and electrical engineering, and digital transformation.

## EDUCATION

**Universitas Gadjah Mada — B.Sc. Electronics & Instrumentation** (International Undergraduate Program)
*Yogyakarta, Indonesia* · Aug 2023 – 2027 (Expected)

- GPA: 3.88 / 4.00
- Coursework: Advanced Control Systems, Industrial Instrumentation, Sensor & Actuator Systems, Digital Signal Processing, Computer Networks, Deep Learning, Computer Vision, Computer Architecture
- Teaching Assistant — Computer Networks Laboratory | Workshop Assistant — Electronics Laboratory

**Waseda University — Thesis Research & Exchange Program,** Computer & Communications Engineering
*Tokyo, Japan* · Aug 2025– Feb 2026

- Awarded competitive international exchange for academic excellence; one of a select cohort from UGM enrolled at a top-tier Japanese research university.
- Coursework: Advanced Intelligent Software, Information Retrieval, Artificial Intelligence, Software Engineering

## EXPERIENCE

**PT Pertamina Hulu Rokan | Duri, Riau, Indonesia** *Jun 2026 – Jul 2026*
Digital Transformation ICES Engineering Intern — Instrumentation, Control, Electrical & SCADA (On-site)

- Engineered an automated predictive maintenance pipeline: fully developed and deployed a robust machine learning workflow to evaluate the expert system, benchmarking traditional ARIMA models against Random Forest, XGBoost, and LSTM across 23 critical pump sensor parameters to enhance early-failure detection.
- Built a Programmatic Reporting & Visualization Workflow: Designed an end-to-end automated reporting system using Python and JavaScript to seamlessly transform raw sensor data into comprehensive HTML dashboards, statistical charts, and presentation-ready PowerPoint decks and PowerBI.
- Conducted In-Depth Statistical Model Evaluation: Led comprehensive R-Squared and error metric analyses on model performance, delivering actionable technical insights to optimize early warning systems and support a hybrid-architecture transition for critical operations at PT Pertamina Hulu Rokan.

**Head of Project Management Department** · *Elins Research Club — UGM Engineering Organisation*
Dec 2024 – Jul 2025 · Yogyakarta, Indonesia

- Directed the full project delivery portfolio of a 50-person, multi-department engineering organisation spanning 4 technical divisions and 8+ concurrent R&D projects, including Battery Management System (BMS) and embedded electronics programs relevant to oilfield technology.
- Deployed a centralised project tracking and reporting infrastructure (Excel, Atlassian Trello) from scratch, achieving real-time milestone visibility across all workstreams and eliminating coordination bottlenecks across divisions.
- Developed and managed a direct team of 10 Project Managers through structured Agile sprint planning and cross-functional frameworks; delivery performance formally recognised by organisational leadership.

**External Relations & Partnerships Manager** · *GAMAFORCE UGM — National Aerospace Research Team*
Jan 2025 – Aug 2025 · Yogyakarta, Indonesia

- Led corporate strategy and multi-stakeholder negotiation for a competitive national autonomous UAV team, securing institutional backing from PERTAMINA Kilang Internasional, PLN, Waskita, BRI, and Buaya Aerotech. Outcomes supported the team's Overall Champion title at Indonesian Flying Robot Contest (KRTI) 2025, Indonesia's national autonomous UAV competition under Ministry of Education.

**Data Engineering Intern (Project-Based)** · *ID/X Partners & Rakamin Academy*
Feb 2024 – Mar 2024 · Jakarta, Indonesia · Remote

- Designed and documented end-to-end ETL pipeline architectures covering data warehouse fundamentals. Applied data visualisation to structure analytical findings for non-technical stakeholders; working knowledge of stored procedures, SQL query scheduling, and fault-tolerant job queue management

## ACADEMIC RESEARCH & TECHNICAL PAPERS

**Fault-Tolerant LoRaWAN Communication System for Mining Early Warning Systems**
*Technical Research Paper · UGM* 2025

- Designed a redundant LoRaWAN gateway topology and adaptive data rate algorithm for a safety-critical IoT communication network in high-attenuation of mining environments.

**Computational Analysis of Transformer Architectures for Barren Land Segmentation Using Earth Observation Data**
*Final Thesis · Waseda University · UGM* 2026

- Benchmarking & evaluating Vision Transformer variants against baselines on remote sensing satellite imagery, with focus on efficiency-accuracy constrained compute, related to satellite-based pipeline monitoring, land-use analysis, and environmental compliance in upstream O&G.

**Advancements in Valve Actuator Technologies for Offshore & Subsea Installations**
*Technical Review · UGM* 2024

- Comparative technical review of hydraulic, pneumatic, electric, and electro-hydraulic actuator systems for extreme subsea environments; concluded electric actuators present the most favourable profile for deep-water IIoT-integrated deployments.

**Gradient Projection Method Investigation in Quadratic Programming for Support Vector Machines**
Research Work · Waseda University 2025

- Deep analysis of Gradient Projection, Active-Set, and Interior-Point methods as optimisation solvers for the QP formulation of SVMs; identified practical advantages in large-scale warm-start optimisation scenarios.

**Analysis of Prefix Front Coding for Data Compression in Distributed Systems**
Technical Analysis · Waseda University 2025

- Analysed algorithmic time complexity, memory overhead, and compression ratios of Prefix Front Coding across varied data distributions; derived optimisation strategies for real-time distributed database and search index applications.

## TECHNICAL SKILLS

**Software & Simulation:** Python, SQL, C++, MATLAB (DSP), Cisco Packet Tracer, Proteus Circuit Simulation, Autodesk EAGLE, LaTeX

**AI / Data:** Deep Learning (CNN, RNN, Vision Transformers), AI/ML Benchmarking, ETL/ELT Pipeline Architecture, Data Warehouse Design, Data Visualisation

**Project Management:** Agile / Scrum, Sprint Planning, Atlassian Trello, Jira, Notion, Cross-functional Coordination, Technical Documentation, Executive Reporting

**Industrial & Field:** Instrumentation & Control Systems, Well Control (L1), SCADA / DCS Awareness, P&ID Reading, LoRaWAN Industrial IoT, Loop Checking, Field Instrument Calibration Principles, Embedded Systems, PCB Design (KiCAD, CADEagle)

**Microsoft Office:** Excel (Advanced — dashboards, pivot tables, reporting), Word, PowerPoint, Teams

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Building AI Solutions Using Advanced Algorithms and Open Source Frameworks — IBM · Apr 2026
- IWCF Level 1 Programme — International Well Control Forum (IWCF) · Issued Jun 2026
- IELTS English Academic Band 7.0 (C1 Level) — IDP IELTS Indonesia · Issued Sep 2024
- Fundamental P&ID (Piping and Instrumentation) Engineering Diagram Drawing Using AutoCAD — Anak Teknik Indonesia
- Electric Industry Operations and Markets — Duke University / Coursera

## AWARDS & HONOURS

- **Overall Champion — Indonesian Flying Robot Contest (KRTI) 2025** — Ministry of Education, Culture, Research & Technology · GAMAFORCE Team · 2025
- **International Exchange Tuition Scholarship — Waseda University** — Competitive merit-based partial scholarship for academic excellence · 2025
- **Delegate, Global Case Competition — Harvard University** — IBM & Equinix strategic acquisition case · IFSA Network & Harvard GSAS Business Club · 2024

## LANGUAGES

**Indonesian:** Native | **English:** Professional Proficiency — IELTS Academic Band 7.0 (C1)